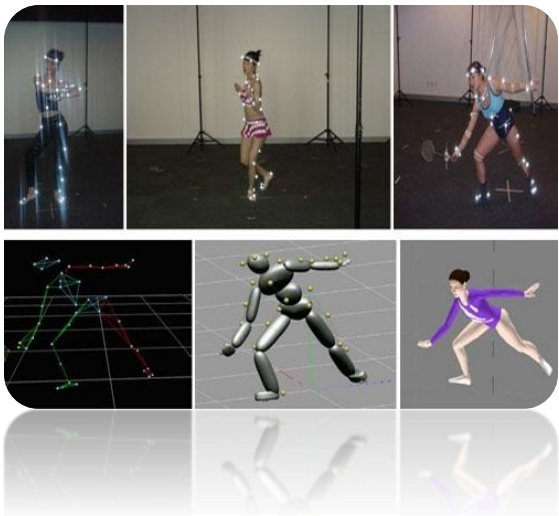




运动捕获(Mocap)

蒋浩

Email: jianghao@ict.ac.cn

运动捕获 (Mocap)



猩球崛起3: 终极之战

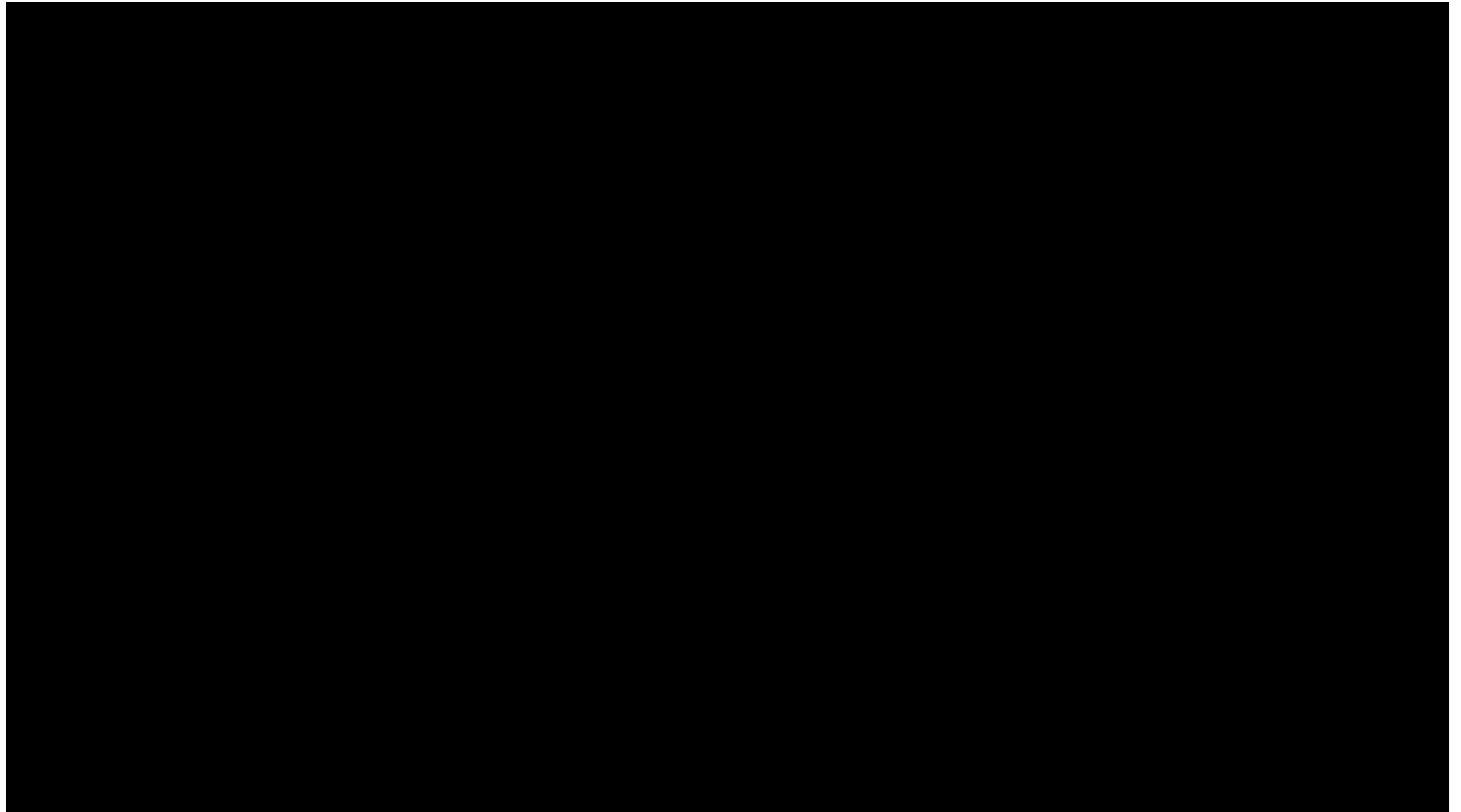
Photo Source: Courtesy of 20th Century Fox – Pictured: Andy Serkis in War for the Planet of the Apes

运动捕获定义(Wiki百科)

- Motion capture is the process of recording the movement of objects or people.
- In filmmaking and video game development, it refers to recording actions of human actors, and using that information to animate digital character models in 2D or 3D computer animation
- When it includes face and fingers or captures subtle expressions, it is often referred to as performance capture(表演捕捉).
- In many fields, motion capture is sometimes called motion tracking, but in filmmaking and games, motion tracking usually refers more to match moving.

运动捕获

- 运动捕获(MoCAP, Motion Capture)采用软硬件系统记录表演者的真实运动信息, 并把动作过程复制到一个虚拟的人或动物上。
- 运动捕获可以获取表演者动作的个性和运动细节, 是生成逼真人体动画最实用、最有效的方法, 因而在影视特技、游戏、动画、机器人控制中被广泛采用。



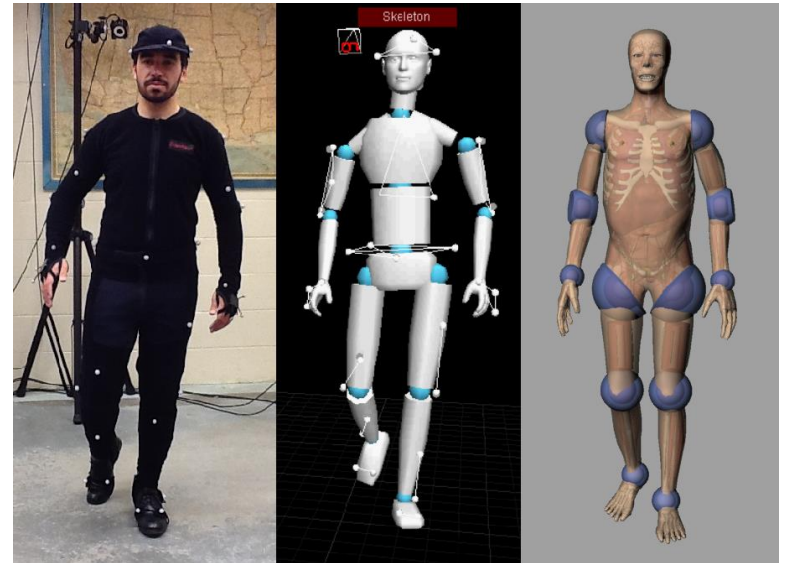
采用运动捕获的游戏

- NBA live
- PGA tour (PGA巡回赛)
- NHL hockey (NHL冰球大赛)
- Legends of Wrestling 2 (摔跤传说2)
- Lords of Everquest (无尽的任务战略版)
- Lord of the Rings
- Tony Hawk Pro Skater (托尼霍克滑板)
- Batman: Dark Tomorrow (蝙蝠侠:黑暗未来)
- Grand Theft Auto (侠盗猎车手)
- Soul Caliber (刀魂)

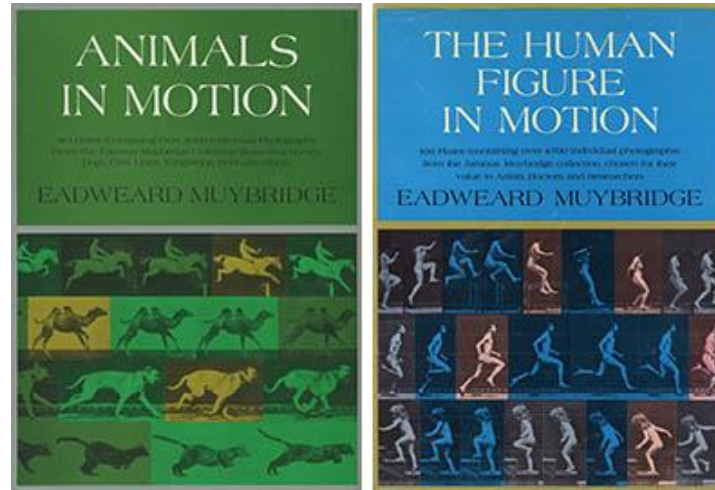


为什么需要运动捕获？

- 正向运动学、逆向运动学
 - 制作物理真实的运动是件非常艰巨的任务
 - 需要大量的专业动画人才
- 把运动进行记录并把结果映射到虚拟角色
 - 更加容易
 - 真实运动

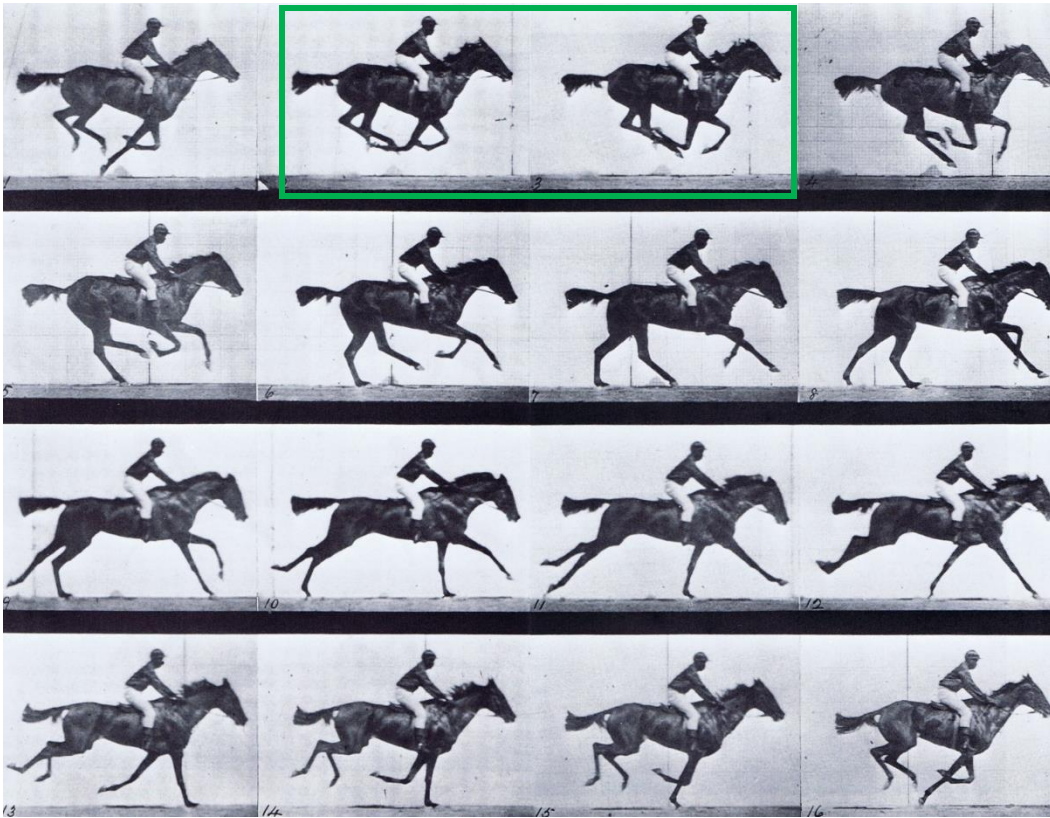


运动捕获历史



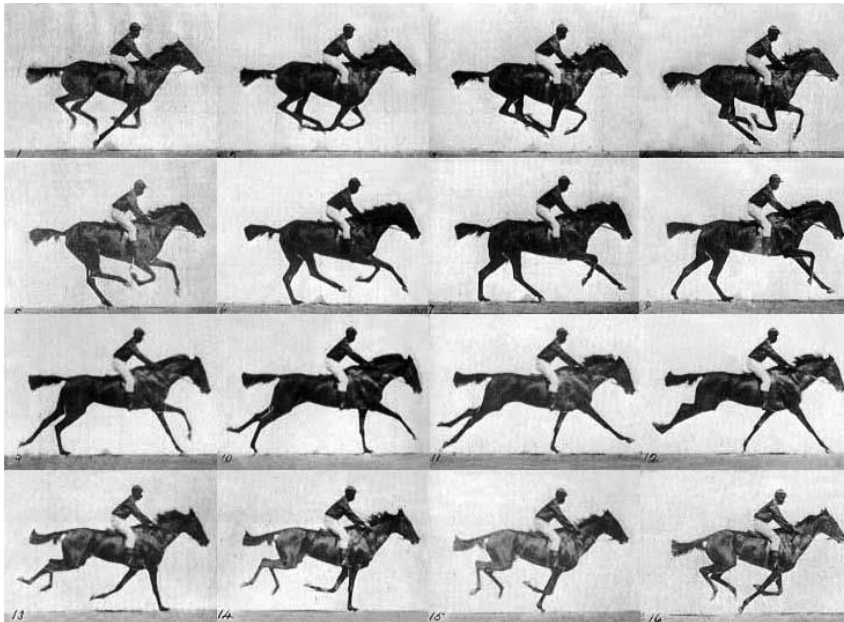
- **Eadweard Muybridge (1830 - 1904)**
- English photographer, pioneered photographic studies of motion and motion-picture projection.
- In 1872 former governor of California Stanford hired Muybridge to prove all four feet of **a horse were off the ground at the same time while trotting (小跑)**.

《Animals in Motion》

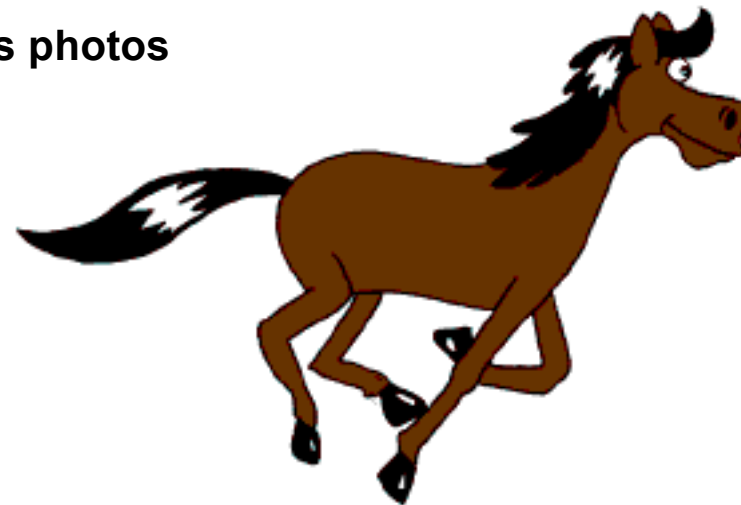


影像描摹(Rotoscoping)

- Allow animators to trace **cartoon character** over photographed frames of live performances
 - Invented by **Max Fleischer in 1915**
 - 2D manual motion capture



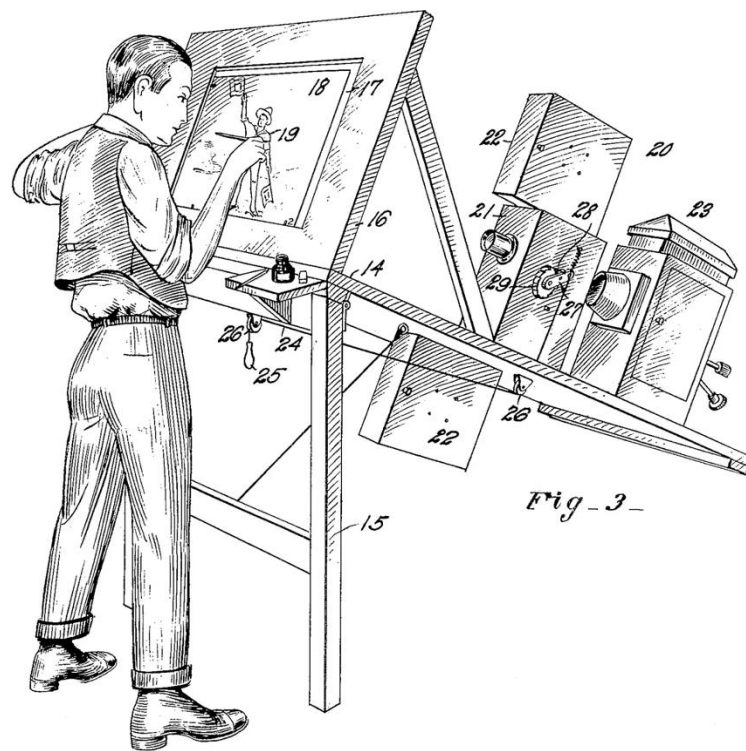
A horse animated by
rotoscoping from
Muybridge's photos



影像描摹(Rotoscoping)

运动捕获方法类似于早期Disney公司制作卡通片《白雪公主》时使用的

“rotoscoping”技术，即动画师根据素材画面，采用手工交互的方式跟踪获取画面主体的运动信息。



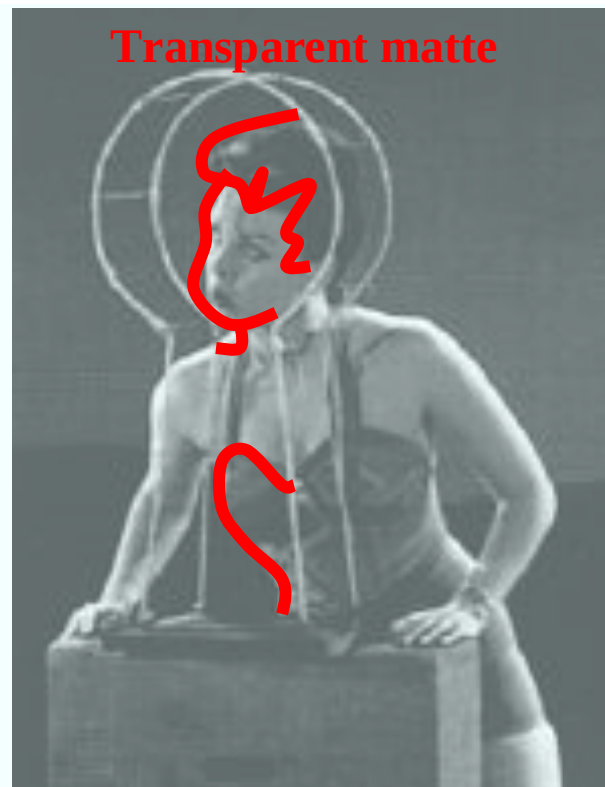
影像描摹(Rotoscoping)

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Captured Video



Tracing



Post Processing

运动捕获系统分类

可看作“**3D Rotoscoping**”: measuring **3D** positions, orientations, velocities or accelerations **automatically**

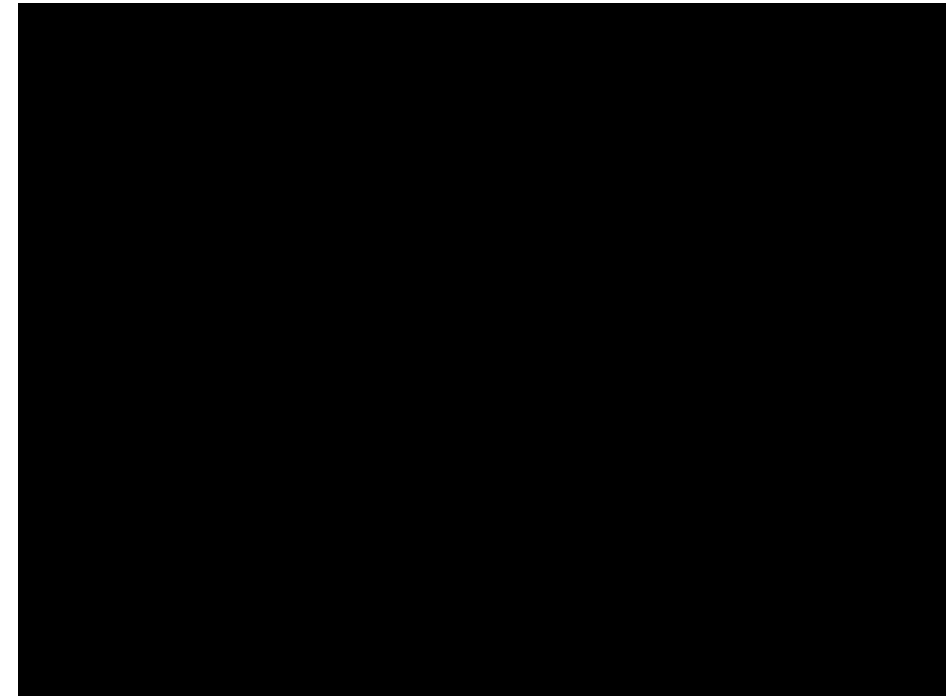
- **电磁**捕获系统(Electromagnetic)
- **机电**捕获系统(Electromechanical)
- **光纤**捕获系统(Fiber optic)
- **光学**捕获系统(Optical)
- **基于视频**的捕获系统(Video-based Mocap)

电磁捕获系统(Electromagnetic)

- Each sensor record 3D position and orientation
- Each sensor placed on joints of moving object
- Full-body motion capture needs **at least 15 sensors**
- Popular system: <http://www.ascension-tech.com/>



军事中的电磁跟踪系统



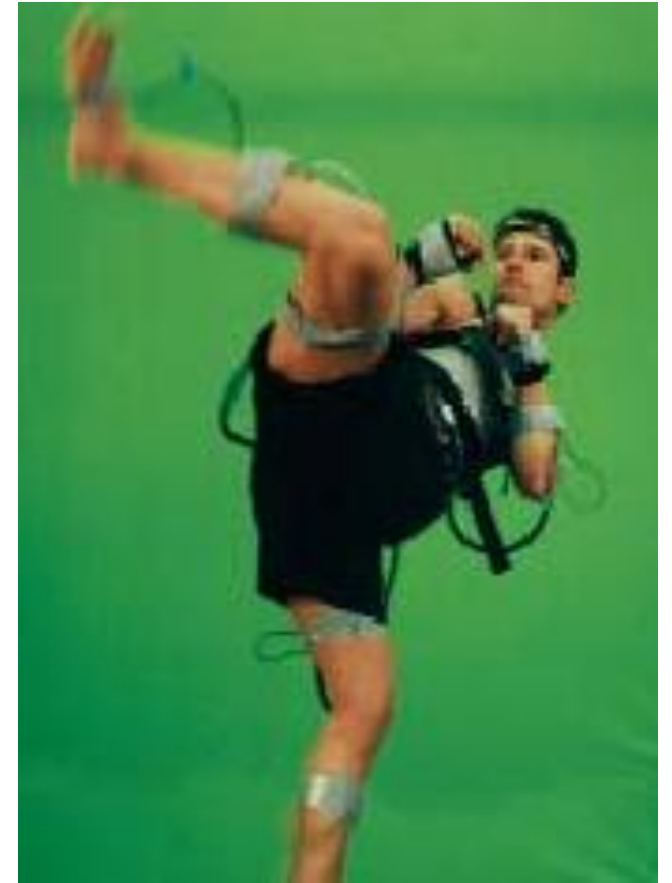
电磁捕获系统(Electromagnetic)

- **Pros**

- measure 3D positions and orientations
- no occlusion problems
- can capture multiple subjects simultaneously

- **Cons**

- magnetic perturbations (metal)
- small capture volume
- cannot capture deformation (facial expression)
- hard to capture small bone movement (finger movement)
- not as accurate as optical mocap systems



机电捕获系统(Electromechanical)

- Each sensor measures 3D orientations
 - including 3D accelerometers(加速度传感器), 3D gyros(陀螺), and 3D magnetometers(磁传感器)
- Each sensor placed on joints of moving object
- Full-body motion capture needs **at least 15 sensors**
- Popular systems: <http://www.xsens.com/>



机电捕获系统(Electromechanical)

- **Pros**

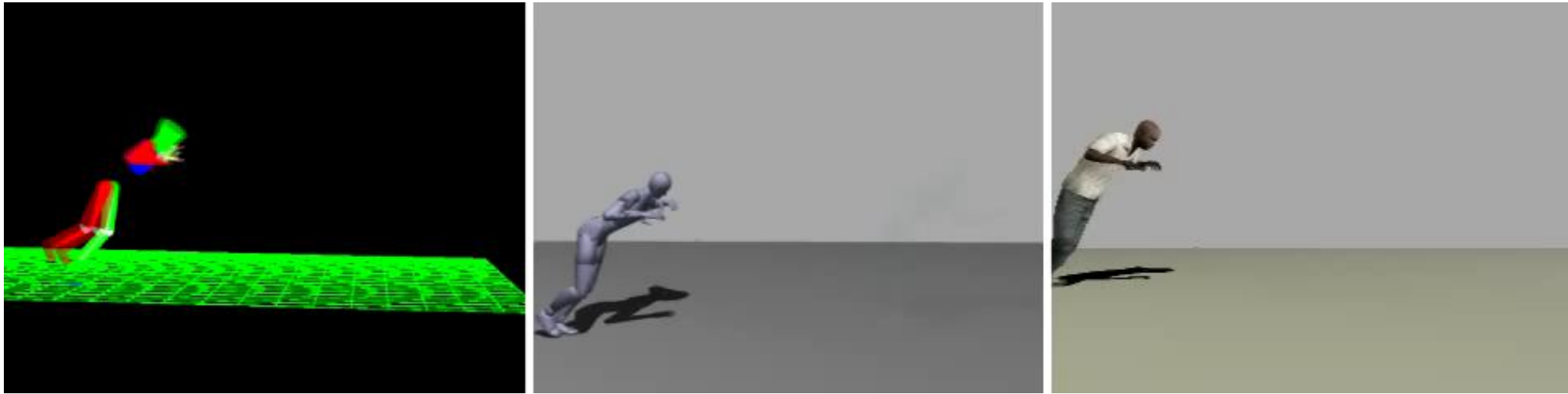
- measure 3D orientations
- no occlusion problems
- can capture multiple subjects simultaneously
- large capture volume
- portable and outdoors capture (e.g. skiing)

- **Cons**

- getting 3D position info is not easy
- the root positions is often measured with ultrasonic position sensors
- cannot capture deformation (facial expression)
- hard to capture small bone movement (finger motion)
- not as accurate as optical mocap system



光纤捕获系统(Fiber optic)



- Measures 3D position and orientation of entire tape
- Binding the tape to the body
- Popular systems: <http://www.measurand.com/>

光纤捕获系统(Fiber optic)

- **Pros**

- measure 3D positions and orientations
- no occlusion problems
- can capture multiple subjects simultaneously

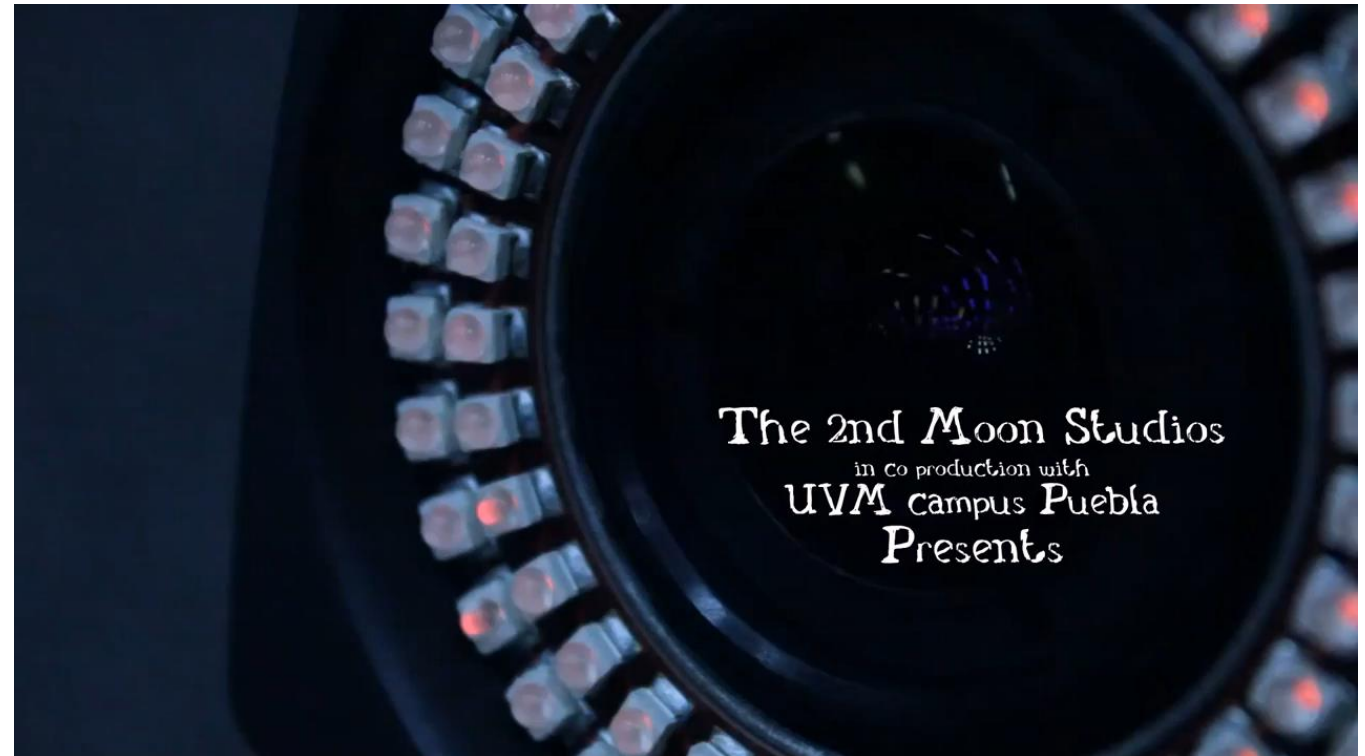
- **Cons**

- small capture volume
- cannot capture deformation (facial expression)
- hard to capture small bone movement (finger movement)
- not as accurate as optical mocap systems



光学捕获系统(Optical)

- Multiple **calibrated cameras** (≥ 8) digitize different views of performance
- Wears retro-reflective markers(反光标记)
- Accurately measures 3D positions of markers
- Vicon mocap system:
<http://www.vicon.com>



光学捕获系统(Optical)

- **Pros**

- measure 3D positions and orientations
- the most accurate capture method
- very high frame rate
- can capture very detailed motion (body, finger, facial deformation, etc.)

- **Cons**

- has occlusion problems
- hard to capture interactions among multiple actors
- limited capture volume
- expensive



基于视频的捕获系统(Video-based Mocap)

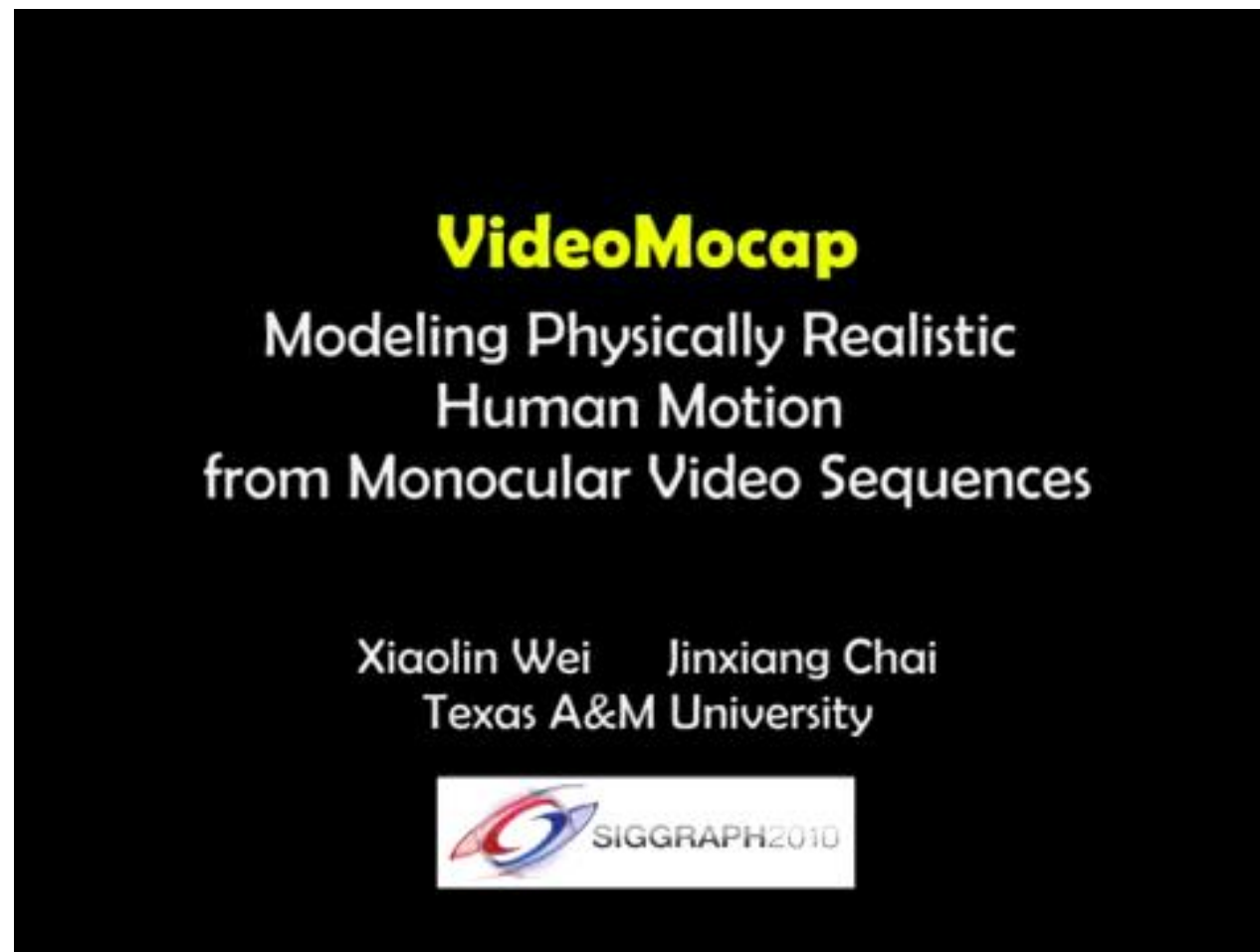
- Capturing 3D performance from single-camera video streams

Pros:

- capturing human motion anytime, anywhere
- very cheap
- zillions of films, sports footage, and internet videos.

Cons:

- not a mature technology
- quality is not as good as other capturing technologies.



近几年活跃方向：基于深度学习的姿态恢复

GLAMR: Global Occlusion-Aware Human Mesh Recovery
with Dynamic Cameras

Ye Yuan¹ Umar Iqbal² Pavlo Molchanov² Kris Kitani¹ Jan Kautz²

¹Carnegie Mellon University ²NVIDIA

 With Narration

运动捕获主要思路 (关节)

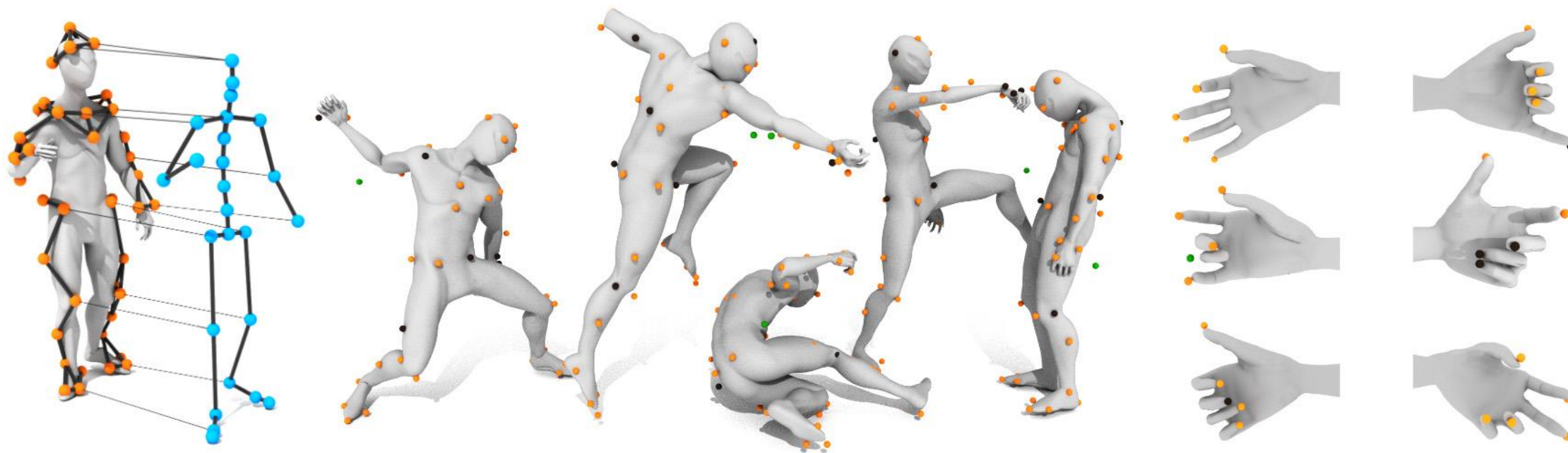
- 跟踪参考点的运动
- 把参考点的信息转化为关节角度
- 用关节角驱动关节模型
- 结合运动轨迹，进行角色的控制



MOCAP流水线（关节）

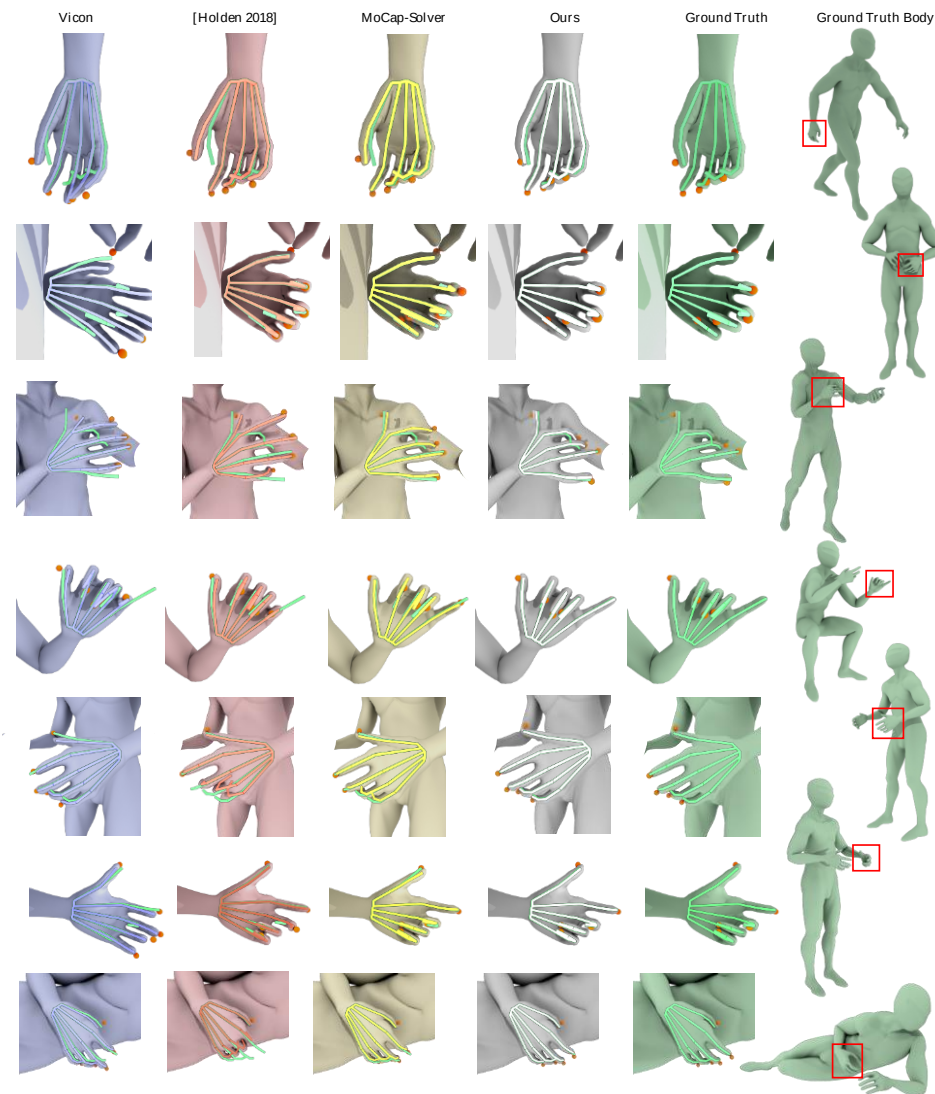
- 规划(Planning)
- 标定(Calibration)
- 捕获(Capture)
- 数据清洗(Marker Cleaning)
- 三维位置重建(3D Position Reconstruction)
- 拟合到骨架(Fitting to the Skeleton)
- 后处理(Post Processing)

Neural Solver for Optical Motion Capture

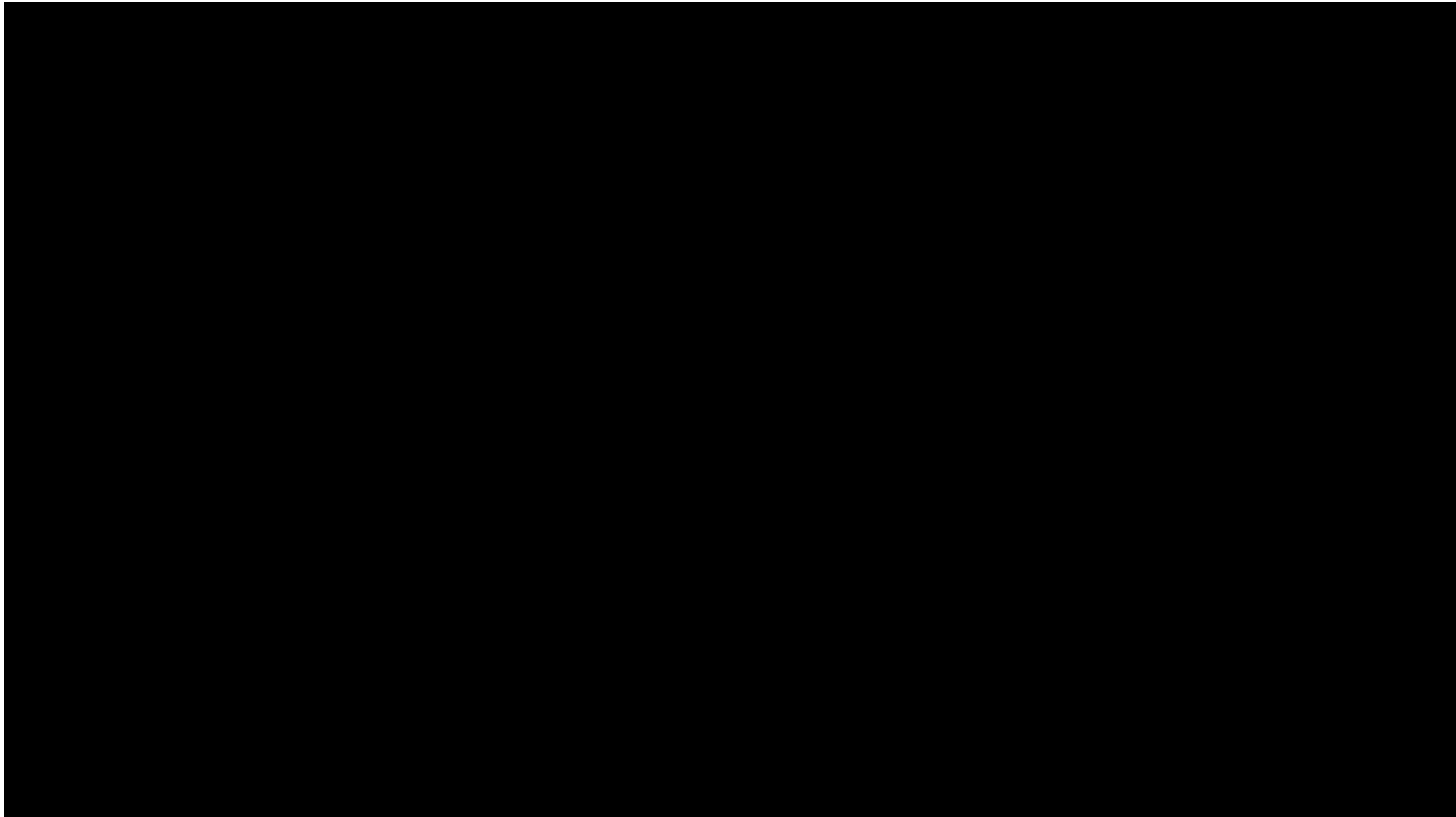


[Xiaoyu Pan](#), [Bowen Zheng](#), Xinwei Jiang, Guanglong Xu, Xianli Gu, Jingxiang Li, Qilong Kou, [He Wang](#), Tianjia Shao, Kun Zhou, [Xiaogang Jin](#), “A Locality-Based Neural Solver for Optical Motion Capture”, **Proceedings of Siggraph Asia'2023, 2023.**

Neural Solver for Optical Motion Capture



Neural Solver for Optical Motion Capture



Motion Capture的利弊

- 优点

- 只要能被捕获，可记录人体运动的所有细节，运动真实

- 缺点

- 不容易进行

- 清洗
- 编辑
- 控制

- 较昂贵



Animators could use more than 750 controls to create Shrek's performance. Some controlled one joint or muscle, others controlled groups of several.

Online Motion Capture Database: <http://mocap.cs.cmu.edu/>

艺术家对MOCAP的评述

Eric Darnell, co-director of Antz



- **The main problem** with motion capture associated with characters has to do with **mass distribution, weight, and exaggeration**. He says that it is impossible for a performer to produce the kind of motion exaggeration that a cartoon character needs, and the mass and weight of the performer almost never looks good when **applied to a character of different proportions**.

艺术家对MOCAP的评述

Richard Chuang, VP at PDI



- The mapping of human motion to a character **with non-human proportions doesn't work**, because the most important things you get out motion capture are the **weight shifts and the subtleties and that balancing act of the human body**. If the proportions change, you throw all that out the door, so you might as well animate it.

艺术家对MOCAP的评述

Godzilla: Karen Goulekas



- The reason that we pulled the plug (终止) on using the motion capture was, very simply, because **the motion we captured from the human actor could not give us the lizard-like motion we were seeking**. The mocap could also not reflect the huge mass of Godzilla either. During our keyframe tests, we found that the Godzilla motion we wanted was one that maintained the sense of huge mass and weight while still moving in a graceful and agile manner. No human actor could give us this result.

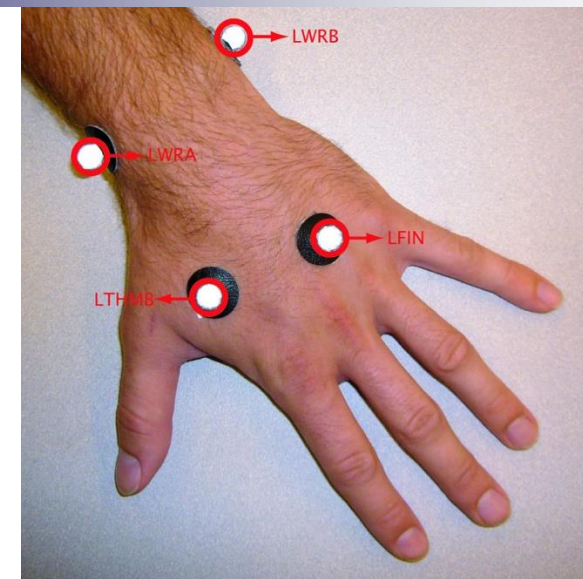
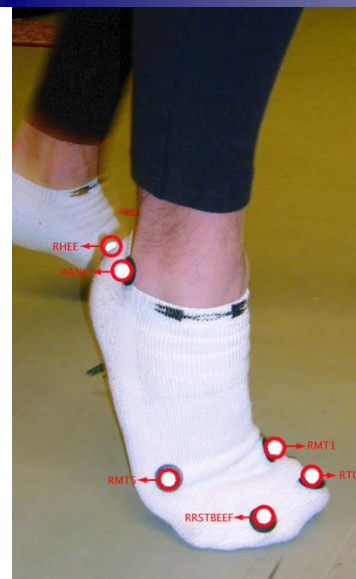
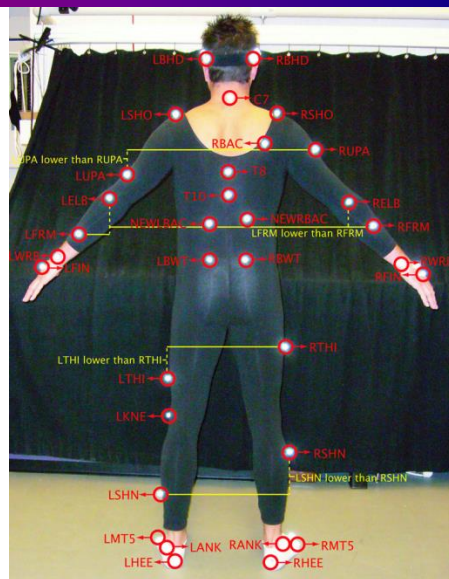
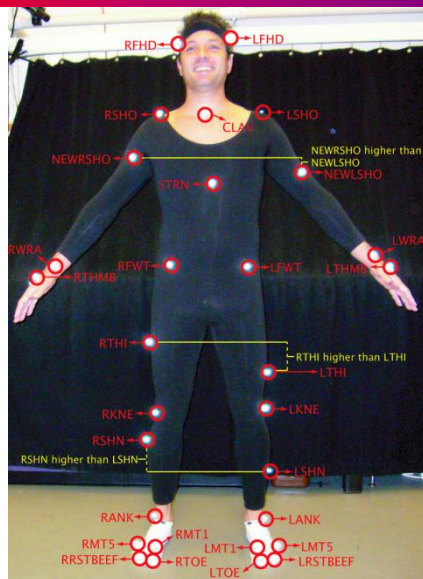
CMU Graphics Lab Motion Capture Database

<http://mocap.cs.cmu.edu/>

- Human Interaction
two subjects
- Interaction with Environment
playground, uneven terrain, ...
- Locomotion
running, walking, ...
- Physical Activities & Sports
basketball, dance, ...
- Situations & Scenarios
common behaviors and expressions, pantomime, ...



CMU Graphics Lab Motion Capture Database



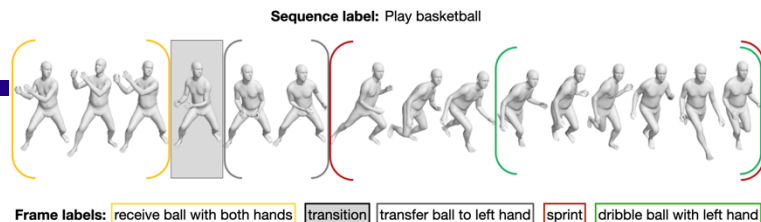
- **How is this data captured?**

We have a **Vicon motion capture system** consisting of **12 infrared MX-40 cameras**, each of which is capable of recording at **120 Hz** with images of 4 megapixel resolution. Motions are captured in a working volume of approximately **3m x 8m**. The capture subject wears **41 markers** and a stylish black garment.

- **运动捕捉数据有多种格式: ASF/AMC, BVH**

一些新的运动捕获数据库

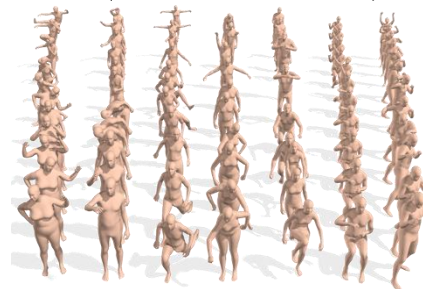
- <https://babel.is.tue.mpg.de/>



CVPR 2021, BABEL: Bodies, Action and Behavior with English Labels (英文标签)

- <https://amass.is.tue.mpg.de/> (smpl格式, 带有mesh)

ICCV 2019, AMASS: Archive of Motion Capture As Surface Shapes

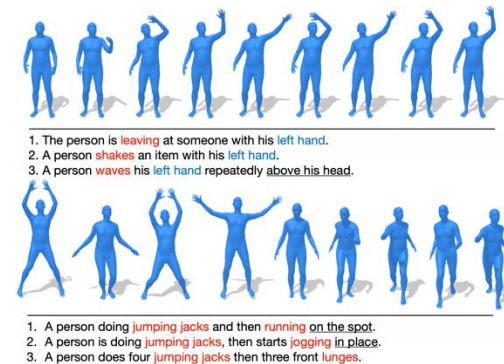


- <https://www.ianxmason.com/100style/> (带100种风格标签)

100STYLE: This is the 100STYLE dataset, it contains over **4 million frames** of motion capture data for **100 different styles of locomotion**.

- <https://github.com/EricGuo5513/HumanML3D>

CVPR 2022, 带文字标签的大型数据集



如何用这些Motion Capture数据?

- 离线(Off-line)
 - 通过滤波、逆向运动学进行处理
 - 建立运动数据库
 - 从中选择
 - 对运动数据进行混合(Blend)
 - 在线(on the fly)修改
- 在线(On-line) (表演动画)
 - 基于演员的实时表演直接驱动虚拟角色

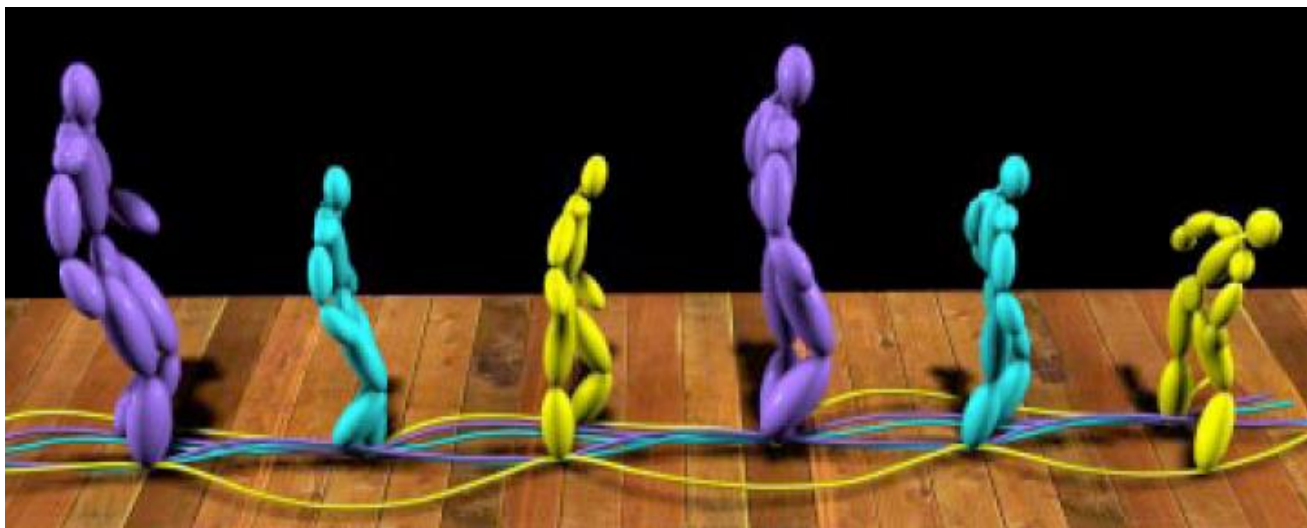
运动捕获数据的处理

- 运动信号处理和运动变形(Motion warping)
- 运动数据的重定向(Motion Retargeting)
 - 把运动映射到不匹配的虚拟角色上，并满足一些重要的约束条件
 - 具有相同/不同的骨骼结构
- 运动数据的合并
 - 把短的运动片段合并成长的运动序列

运动数据的信号处理

- 运动信号处理
 - 考虑信号频率如何刻画各种运动
 - 低频部分：表示基本运动(如走路)
 - 高频部分：特质运动(跛行)
- 运动的操作/编辑/变形
 - 对运动信号进行变形，使得满足用户指定的一些约束条件
- 把运动看成多维信号
- 低通滤波
 - 去噪
- 高通滤波
 - 风格的变换
- 高频信息可能是运动细节，而非噪声
- 通过滤波来进行运动的修改并非易事
 - 物理约束 (关节约束, 地面接触)?
 - 运动的自然性?

运动重定向



Michael Gleicher,
“Retargetting motion to
new characters ”,
SIGGRAPH '98

- 把捕获的运动数据映射到**不匹配的虚拟角色**上，并满足一些重要的约束
- **约束（运动所应满足的性质）**：
 - 避免脚穿透地板,避免自身穿透, 避免走路时的打滑;
- 生成一个与原始运动**尽可能接近的新的运动**，并强制满足约束条件;
- 新的运动可看成是一个时空(space-time)、非线性约束的优化问题。

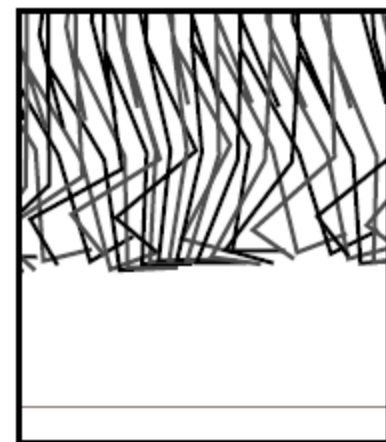
运动重定向

- $m(t) = m_o(t) + d(t)$
 - $m_o(t)$: 原始运动 ; $m(t)$: 重定向的运动
 - $d(t)$: 两个运动之间的距离
- 如果只是简单重用原始的运动, 则目标角色**与其它物体之间的相互作用会失败**。
- 把运动重定向问题看成一个带约束的优化问题:
 - 找到运动 $m(t)$, 满足函数 $f(m(t)) \diamond c$
 - $f(\cdot)$: 约束方程, c : 常数
 - $\diamond \in \{\leq, \geq, =\}$
 - 最小化目标函数: $g(m) = \int_t (m(t) - m_o(t))^2 = \int_t d(t)^2$

原始运动



Naïve重定向运动



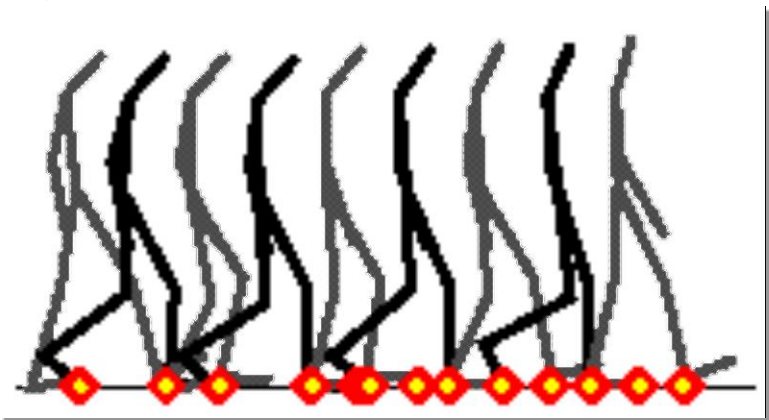
约束例子：Footskate

- **Footskate (打滑)**：是一种经常会发生的不真实现象；
- **Penetrations (贯穿)**：是一种经常会发生的不真实现象；



运动重定向

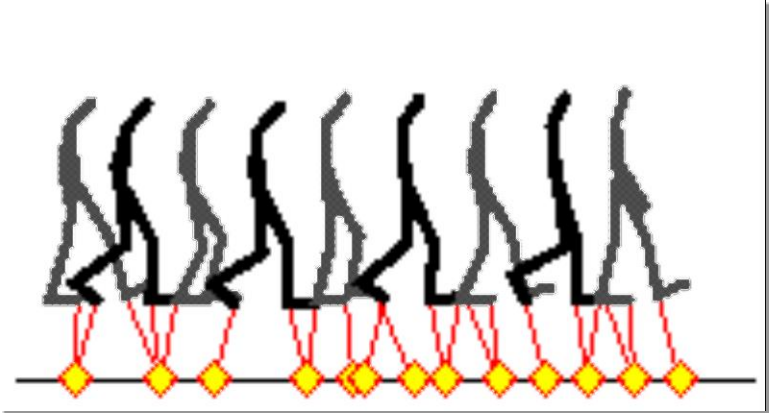
- 定义约束



- 加入平移偏移 (近似解)



- 应用到新的角色



- 解非线性约束优化问题



把运动曲线应用于一个新的角色

- 定义运动的约束条件
- 对于整个运动，求解约束优化问题



运动重定向动画演示



Physics-Based Motion
Retargeting from Sparse Inputs



Physics-based Motion Retargeting from Sparse Inputs, SCA 2023

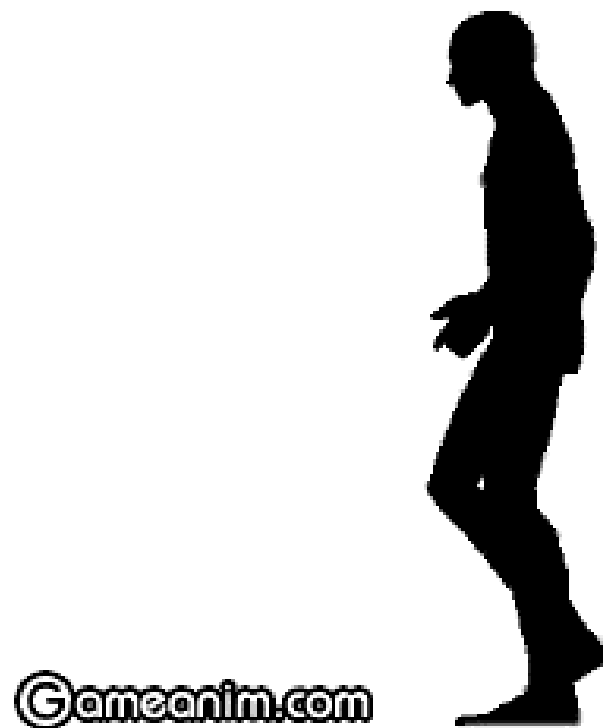
Daniele Reda, Jungdam Won, Yuting Ye, Michiel van de Panne, Alexander Winkler

运动混合(Motion Blending)

- 把两段或多段运动加在一起
 - 实际上是插值

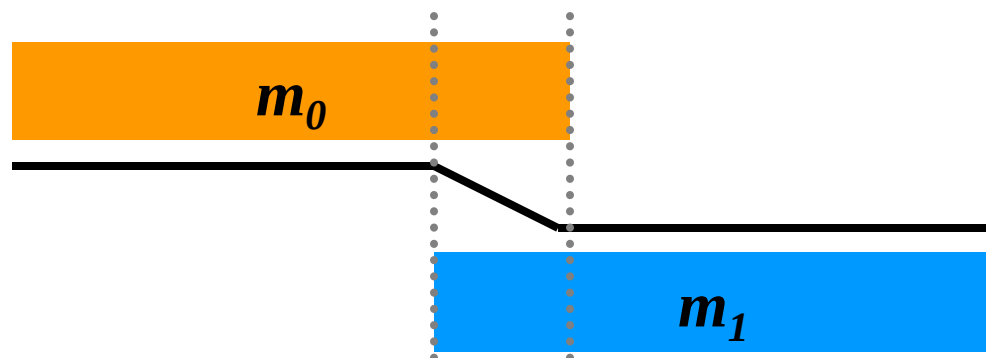
$$m(t) = a m_0(t) + (1-a) m_1(t)$$

这是一个逐帧的操作

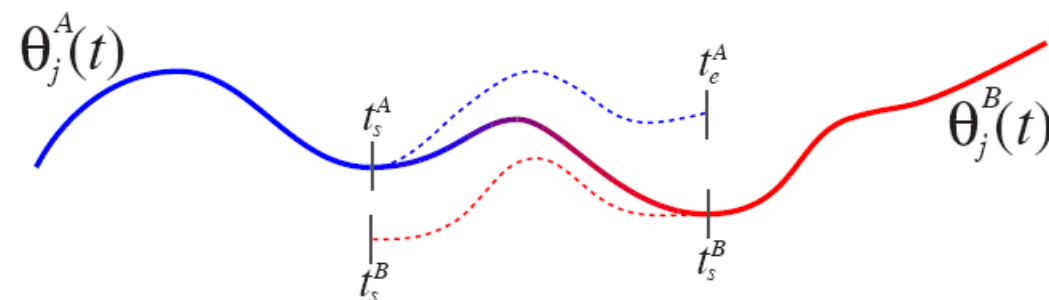


Blend Speed

运动过渡(Motion Transition)

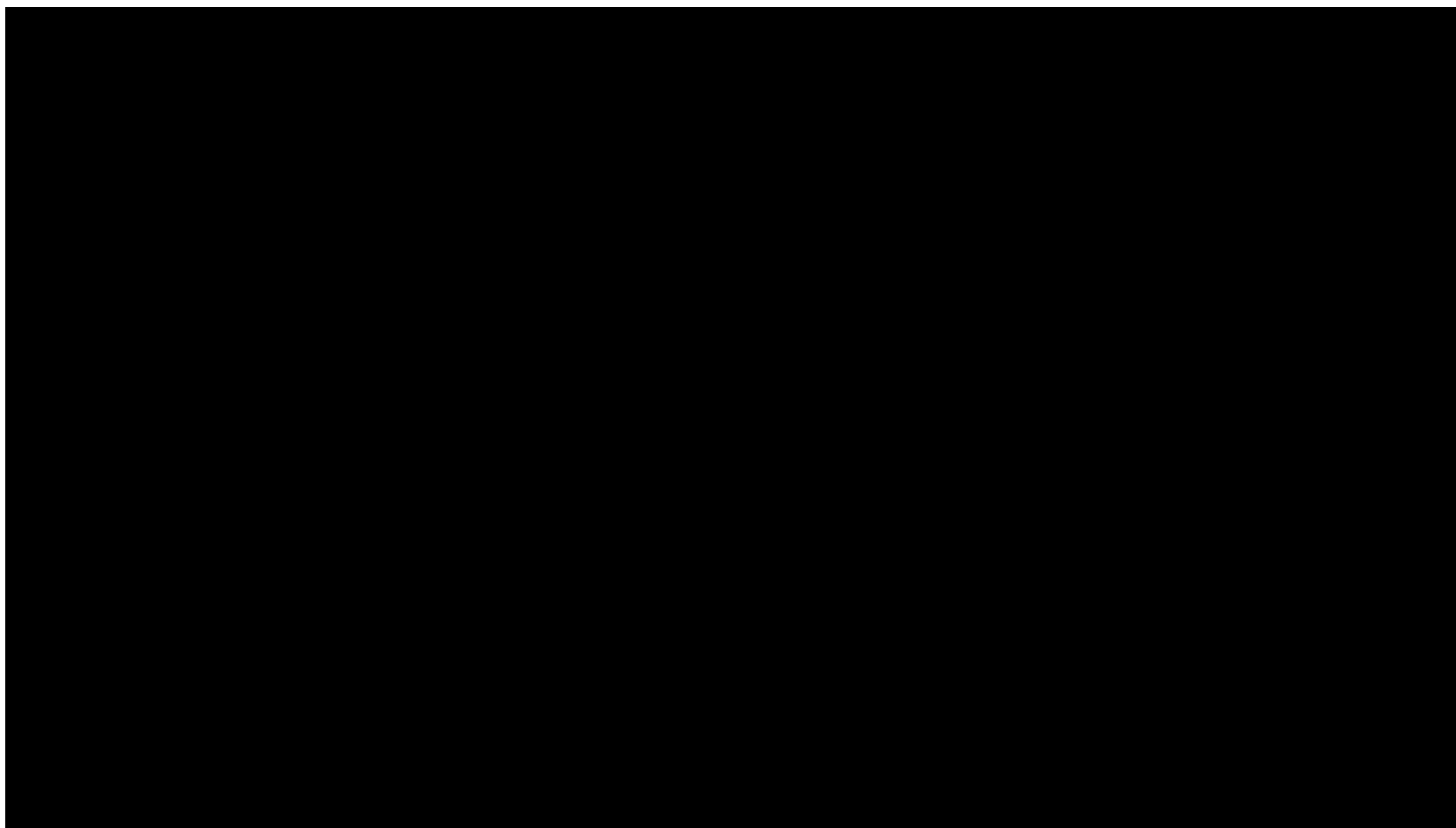


- 运动捕获得到的通常是小的运动片段
- 需要把他们连接起来
- 如果运动是相近的，则很容易



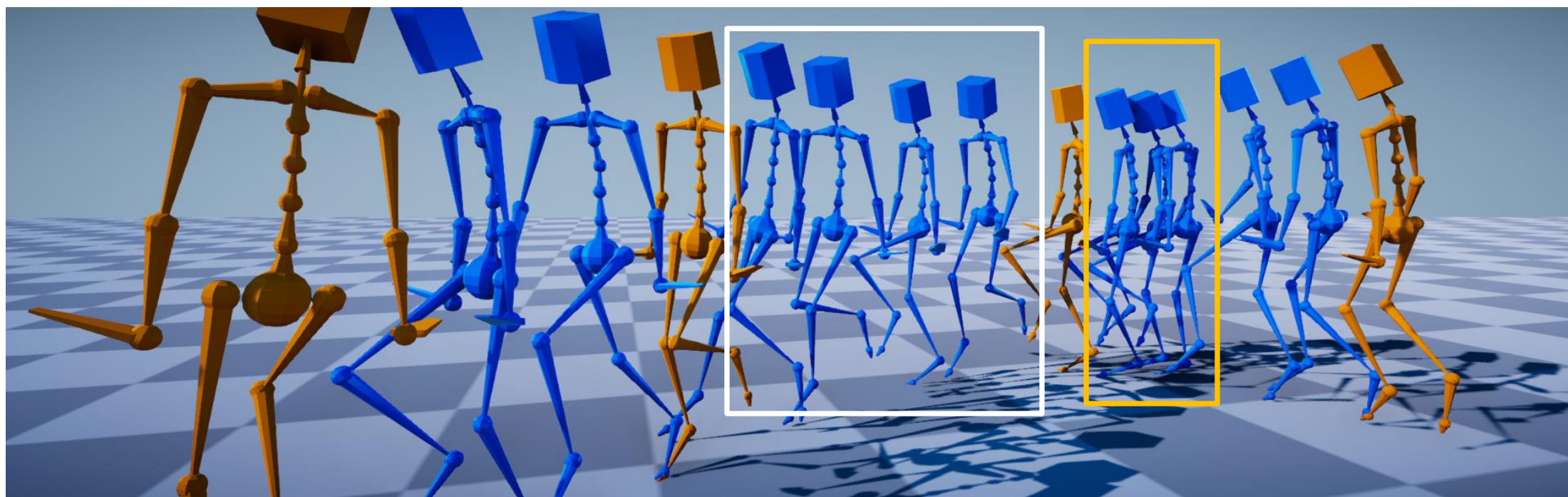
- 如果运动相差较远，则比较困难

运动过渡(Motion Transition)



Xiangjun Tang, He Wang, Bo Hu, Xu Gong, Ruifan Yi, Qilong Kou, Xiaogang Jin, “Real-time Controllable Motion Transition for Characters”, ACM Transactions on Graphics (Proc. Siggraph 2022), 2022, 41(4): Article 137.

带风格的运动过渡(Motion Transition)



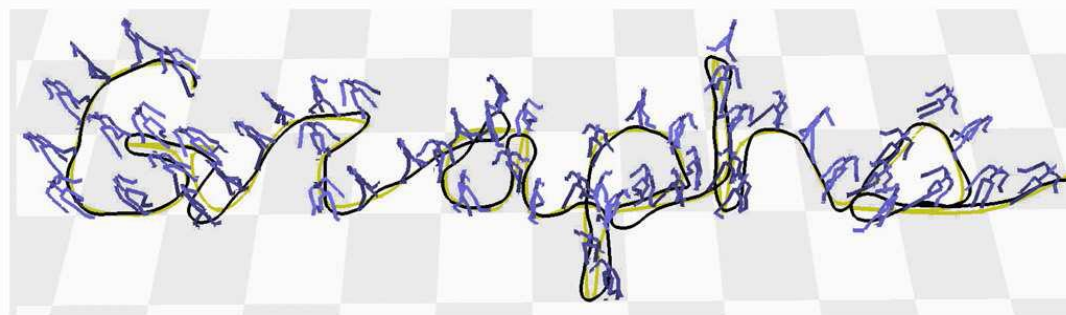
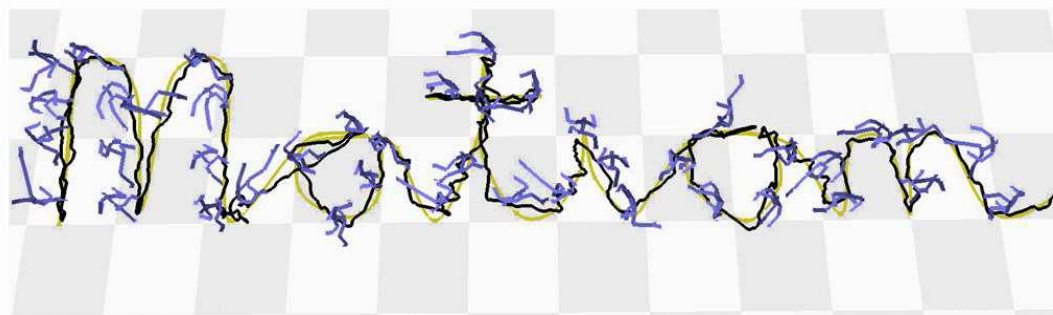
Xiangjun Tang, Linjun Wu, He Wang, Bo Hu, Xu Gong, Yuchen Liao, Songnan Li, Qilong Kou, Xiaogang Jin: RSMT: Real-time Stylized Motion Transition for Characters. *Proc. SIGGRAPH 2023*: 38:1-38:10

Real-time Stylized Motion Transition for Characters

Anonymous author(s)

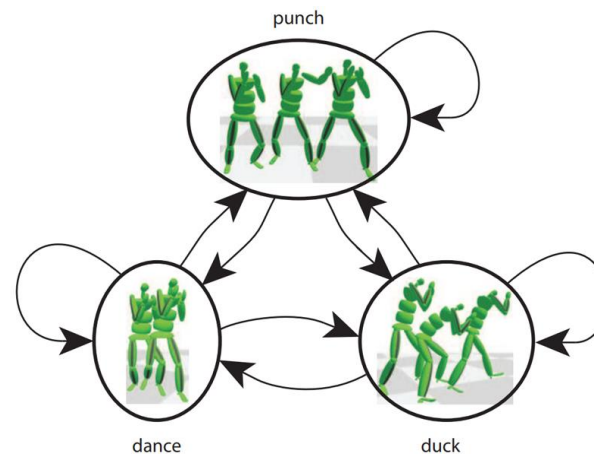
Submission ID: papers_434

运动图(Motion Graph)



把短的运动串联，生成长的运动序列。

检测动画序列中每一帧的相似性，根据阈值找到动画序列中相似度最接近的的动画帧，并生成过渡动画。



Lucas Kovar, Michael Gleicher, Frédéric H. Pighin: Motion graphs. ACM Trans. Graph. 21 (3): 473-482 (2002)

应用实例



《加勒比海盗3》



《指环王》系列电影中的古鲁姆的运动

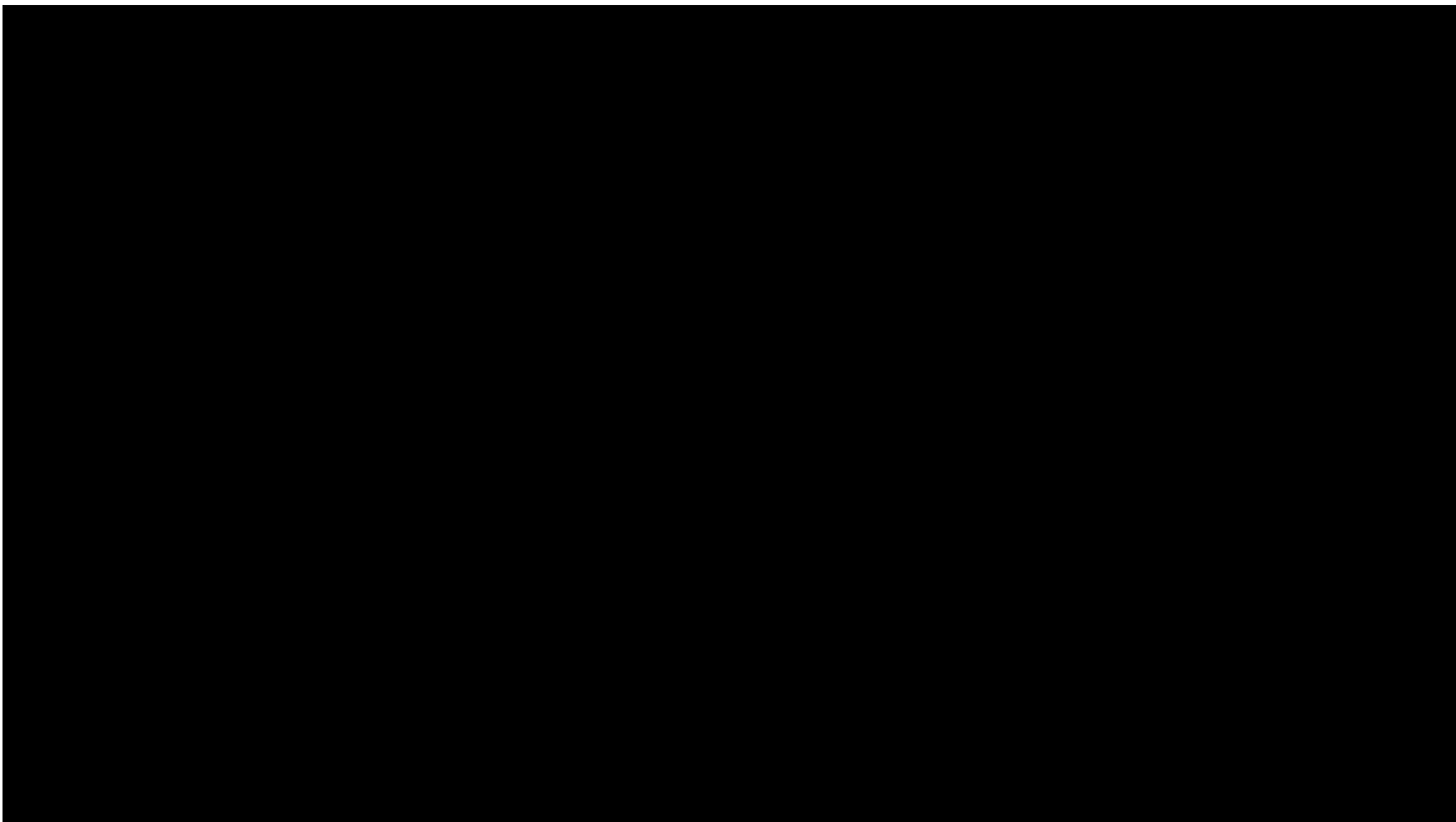
应用实例



表演者

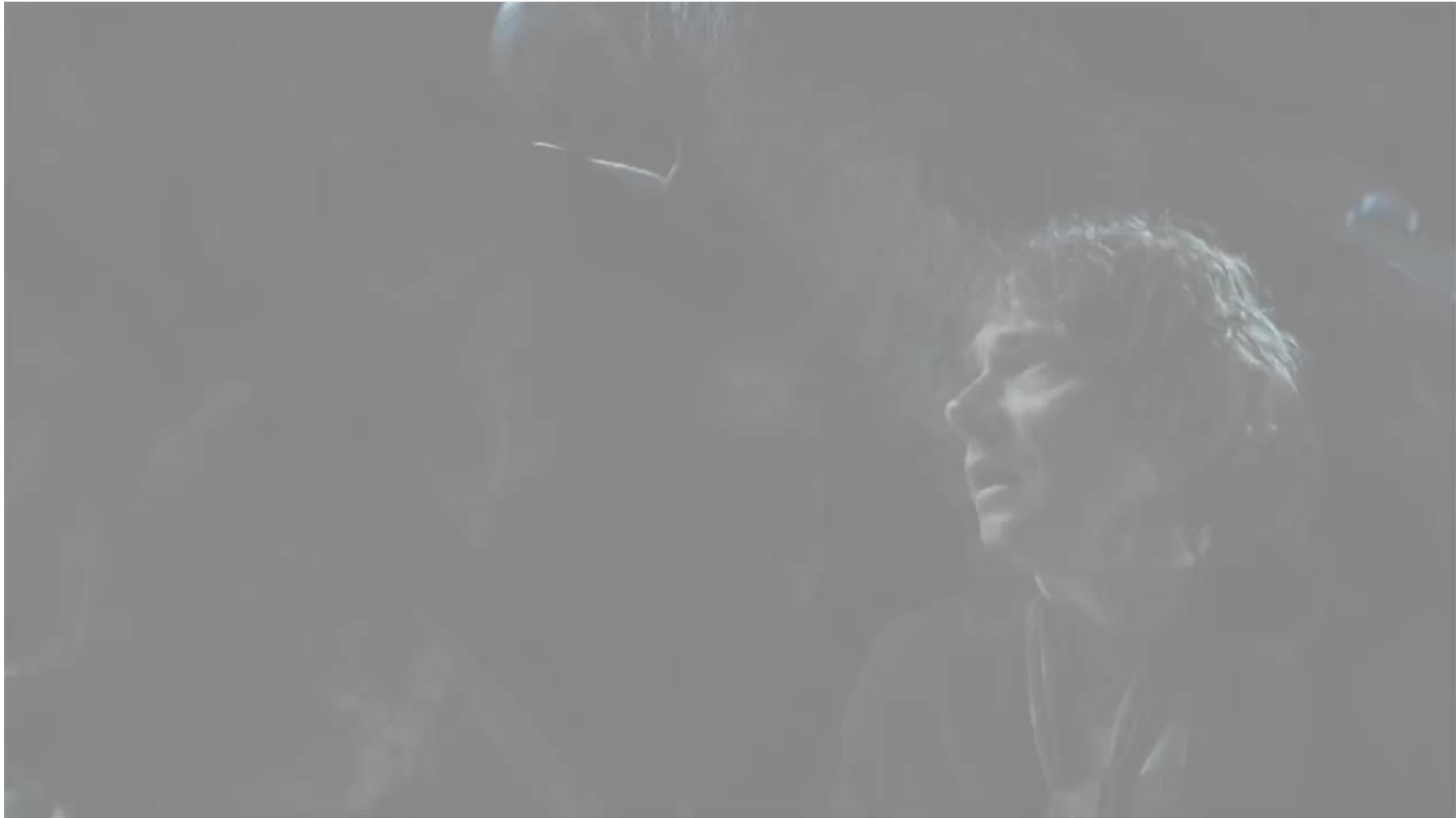


树怪



02:16:00

Creating Gollum



总结

- 运动捕获
- 运动重定向
- 运动混合与过渡

The End